Pranjali A. Chaudhari

+91-7756068761 | pranjalichaudhari0408@gmail.com | linkedin.com/in/pranjali-a-chaudhari | github.com/Pranjali0408 | Pune, India

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP and Generative AI. Developed practical AI solutions including a multimodal healthcare screening system, RAG-based chatbot and LLM security framework using Python, PyTorch and Flask. Skilled in model development, data analysis, deployment and building end-to-end AI applications. Seeking opportunities in AI/ML, Data Science and Generative AI.

Technical Skills

Languages: Python, SQL, Java
AI / ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Generative AI, Large Language Models, Prompt Engineering, Explainable AI (XAI), AI Safety, Federated Learning
Data Science: Pandas, NumPy, Scikit-learn, EDA, Feature Engineering, Data Visualization, Model Training & Evaluation
Tools & Frameworks: PyTorch, Flask, Power BI, Git, Jupyter Notebook, Google Colab, VS Code, SQLite, Matplotlib

Experience

AI / Data Science Intern — Aug 2025 – Nov 2025
DATASMITH AI Solutions — *Pune*

- Trained on end-to-end ML workflows including data preprocessing, feature engineering and model evaluation using Python, Pandas, NumPy and Scikit-learn.
- Applied EDA techniques on structured datasets to extract meaningful insights and support data-driven decisions.
- Gained hands-on exposure to machine learning model development, validation and performance benchmarking.
- Explored real-world AI/ML use cases through structured assignments and industry-oriented case studies.

Projects

CervixAI — Multimodal & Explainable AI for Cervical Cancer Screening | *PyTorch, XAI, Flask* | GitHub — 2025-26

- Designed and implemented a multimodal AI framework integrating image classification and clinical risk assessment for cervical cancer screening support.
- Built a fine-tuned ResNet50, Clinical MLP and Attention-Based Fusion architecture achieving **97.22% CNN accuracy** and **98.80% multimodal classification accuracy**.
- Integrated SHAP and Grad-CAM for model explainability, along with image quality validation and out-of-distribution detection for robustness.
- Developed and deployed a Flask-based web application with SQLite integration and automated PDF report generation.

Adaptive Risk-Aware Multi-Layer Defense Framework for Securing LLMs | *LLM Security, NLP, Python* — 2025-26

- Designed a multi-layer defense framework to protect Large Language Models against adversarial prompt attacks, prompt injection and malicious inputs.
- Implemented adaptive risk scoring, threat detection and input sanitization mechanisms achieving **93.3% defense accuracy** in experimental evaluation.
- Improved LLM robustness through dynamic security policies and risk-aware response filtering.

DharmaNiti — RAG-Based AI Chatbot for Bhagavad Gita & Chanakya Niti | GitHub — 2024-25

- Built a Retrieval-Augmented Generation (RAG) based chatbot leveraging Bhagavad Gita and Chanakya Niti as domain-specific knowledge sources.
- Implemented prompt engineering and semantic retrieval techniques to improve contextual accuracy and reduce hallucinations.
- Designed a domain-specific knowledge retrieval pipeline to generate context-aware and reliable responses.

Packaging Industry Data Analysis & Visualization | *Power BI, Pandas, Data Analytics* — 2023

- Performed data cleaning, preprocessing and exploratory data analysis on real-world packaging industry datasets.
- Designed interactive Power BI dashboards for KPI monitoring, production analysis and defect trend visualization to support data-driven decision making.

Education

Marathwada Mitra Mandal College of Engineering (MMCOE) — 2024 – Present
M.Tech in Data Science — *CGPA: 9.1 / 10*

- *Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Big Data Analytics, Blockchain Technology*

BRACT's Vishwakarma Institute of Information Technology (VIIT) — 2019 – 2023
B.Tech in Mechanical Engineering — *CGPA: 8.51 / 10*

Maharashtra State Board — 2019
HSC (Class XII) — *84.92%*

Maharashtra State Board — 2017
SSC (Class X) — *93.20%*

Research & Publications

A Comprehensive Review on Multimodal and Explainable Deep Learning Approaches for Cervical Cancer Detection | *IEEE ETFI 2026*

- Reviewed 20+ studies on Deep Learning, Computer Vision and Explainable AI for cervical cancer detection.
- *Published in IEEE Conference Proceedings (Scopus Indexed).*

Evaluating Security and Robustness of Instruction-Tuned LLMs Against Adversarial Prompt Attacks | *ICAIS 2026*

- Analyzed adversarial prompt attacks and evaluated robustness of instruction-tuned LLMs.
- *Published in Springer Conference Proceedings (WoS Indexed).*

TMDL: A Trustworthy Multimodal Deep Learning Approach for Cervical Cancer Screening | *Journal Under Review*

- Proposed multimodal cervical cancer screening framework achieving 98.80% classification accuracy.

Adaptive Risk-Aware Multi-Layer Defense Framework for Securing Large Language Models | *Journal Under Review*

- Developed risk-aware LLM defense framework achieving 93.3% defense accuracy.

Certifications

Generative AI Fundamentals — Databricks

Python for Data Science & Machine Learning — Udemy

Python for Everybody — Coursera

Full Stack Web Development Bootcamp — Udemy

Achievements

- Published research papers in **IEEE Scopus Indexed** and **Springer Web of Science Indexed** conference proceedings.
- Achieved **98.80% classification accuracy** in a multimodal AI-based cervical cancer screening system.
- Successfully submitted M.Tech thesis on **Multimodal and Explainable AI for Cervical Cancer Screening**.
- Maintained **9.1/10 CGPA** in M.Tech Data Science.
- Secured **98/100** in Cyber Security Coursework.